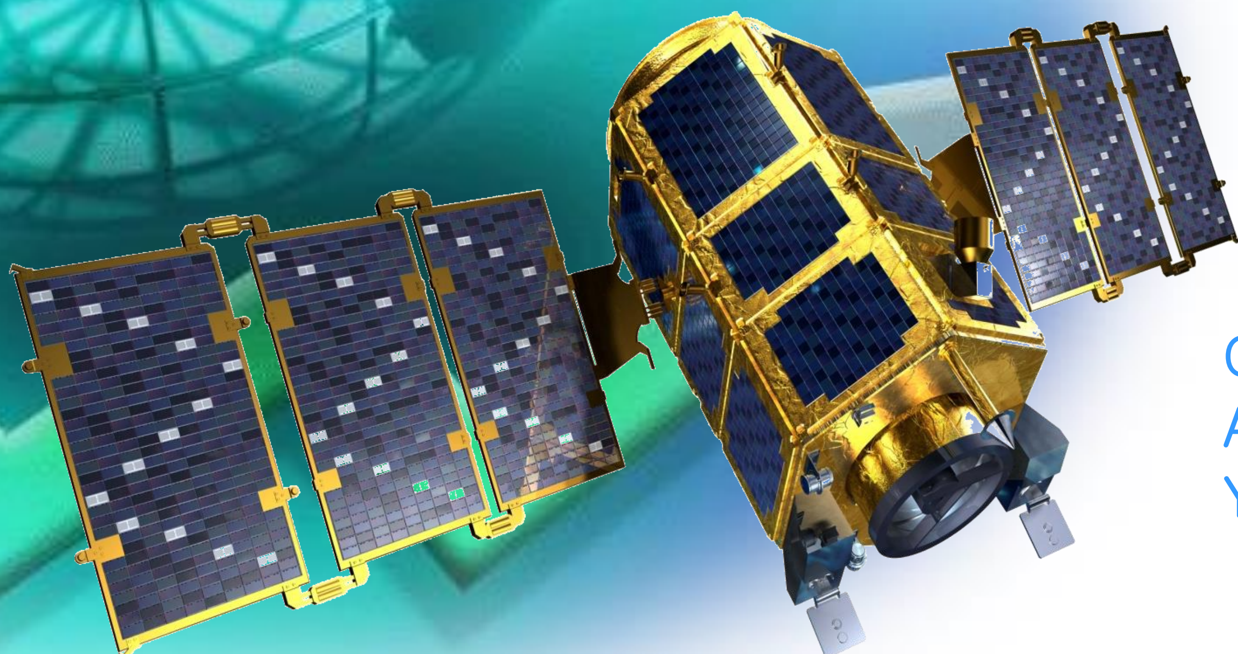
Fall 2011

Spacecraft Systems



Chandeok Park
Astrodynamics & Control Lab.
Yonsei University

Chapter 10
SPACECRAFT DESIGN & SIZING

10.1 Requirements, Constraints, and the Design Process

◆ Overview of S/C Design and Sizing (Table 10-1)

TABLE 10-1. Overview of Spacecraft Design and Sizing. The process is highly iterative, normally requiring several cycles through the table even for preliminary designs.

Step	References
1. Prepare list of design requirements and constraints	Sec. 10.1
2. Select preliminary spacecraft design approach and overall configuration based on the above list	Sec. 10.2
3. Establish budgets for spacecraft propellant, power, and weight	Sec. 10.3
4. Develop preliminary subsystem designs	Sec. 10.4
5. Develop baseline spacecraft configuration	Secs. 10.4,10.5
6. Iterate, negotiate, and update requirements, constraints, design budgets	Steps 1 to 5

10.1 Requirements, Constraints, and the Design Process

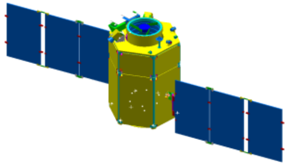
- S/C design process begins by developing baseline requirements and constraints.

TABLE 10-3. Principal Requirements and Constraints for Spacecraft Design. These parameters typically drive the design of a baseline system.

Requirements and Constraints	Information Needed	Reference
Mission:		Chaps. 1, 2
Operations Concept	Type, mission approach	Sec. 1.4
Spacecraft Life & Reliability	Mission duration, success criteria	Secs. 1.4, 10.5, 19.2
Comm Architecture	Command, control, comm approach	Sec. 13.1
Security	Level, requirements	Secs. 13.1, 15.4
Programmatic Constraints	Cost and schedule profiles	Chaps. 1, 20
Payload:		Chaps. 9, 13
Physical Parameters	Size, weight, shape, power	Secs. 9.5, 13.4, 13.5
Operations	Duty cycle, data rates, fields of view	Secs. 9.5, 13.2
Pointing	Reference, accuracy, stability	Secs. 5.4, 9.3, 11.1
Slewing	Magnitude, frequency	Secs. 9.5, 11.1
Environment	Max and min temperatures, cleanliness	Sec. 9.5

10.1 Requirements, Constraints, and the Design Process

Orbit:		Chaps. 6, 7
Defining Parameters	Altitude, inclination, eccentricity	Secs. 7.4, 7.5
Eclipses	Maximum duration, frequency	Sec. 5.1
Lighting Conditions	Sun angle and viewing conditions	Secs. 5.1, 5.2
Maneuvers	Size, frequency	Secs. 6.5, 7.3
Environment:		Chap. 8
Radiation Dosage	Average, peak	Secs. 8.1, 8.2
Particles & Meteoroids	Size, density	Secs. 8.1, 21.2
Space Debris	Density, probability of impact	Sec. 21.2
Hostile Environment	Type, level of threat	Sec. 8.2
Launch:		Chap. 18
Launch Strategy	Single, dual; dedicated, shared; use of upper kick stage	Secs. 18.1, 18.2
Boosted Weight	Launch capabilities	Secs. 18.1, 18.2
Envelope	Size, shape	Sec. 18.3
Environments	g's, vibration, acoustics, temperature	Sec. 18.3
Interfaces	Electrical and mechanical	Sec. 18.3
Launch Sites	Locations, allowed launch azimuths	Sec. 18.1
Ground-System Interface:		Chaps. 14, 15
Degree of Autonomy	Required autonomous operations	Secs. 15.4, 16.1
Ground Stations	Number, locations, performance	Secs. 15.1, 15.5
Space Links	Space-to-space link, performance	Secs. 13.3, 13.4



10.1 Requirements, Constraints, and the Design Process

◆ Mission

- Operations Concept
  - Broad Objectives (Type, Mission Approach) ⇒ Requirements and Constraints ⇒ Operational Concepts ⇒ S/C Orbit, Subsystems
- Spacecraft Life & Reliability
  - Mission Duration ⇒ S/C Life
  - Success Criteria ⇒ S/C Reliability (the probability that a device will function without failure over a specified time period or amount of usage)
- Communication Architecture
  - Command, Control, Comm. Approach ⇒ Comm. Architecture (Network of S/C and Ground Stations) ⇒ S/C Comm., C&DH
- Security
  - Comm. Security ⇒ Comm. Subsystem
- Programmatic Constraints
  - Cost ⇒ S/C cost
  - Schedule Profiles

10.1 Requirements, Constraints, and the Design Process

Payload

- Physical Parameters
  - Size, Weight, Shape, Power (we can understand them in the early design phases) ⇒ S/C physical parameters
- Operations
  - Duty Cycle, Data Rates, Field of View ⇒ S/C Subsystems (Comm., ADCS, C&DH, …)
- Pointing (orientating payload to a target)
  - Reference, Accuracy, Stability ⇒ S/C ADCS
- Slewing
  - Magnitude, Frequency ⇒ S/C ADCS, Propulsion
- Environment
  - Temperature, Cleanliness ⇒ S/C Thermal , Power

10.1 Requirements, Constraints, and the Design Process

◆ Orbit

- Defining Parameters
  - Altitude, Inclination, Eccentricity ⇒ All Subsystems
- Eclipses
  - Duration, Frequency ⇒ S/C Thermal, Power Structure
- Lighting Conditions
  - Sun Angle, Viewing Condition ⇒ S/C Power
- Maneuvers
  - Size, Frequency ⇒ S/C propulsion, ADCS

10.1 Requirements, Constraints, and the Design Process

Environment

- Radiation Dosage
  - Average, Peak ⇒ S/C Structure, Power ⇒ S/C Life
- Particle & Meteoroids
  - Size, Density ⇒ S/C Structure
- Space Debris
  - Density, Probability of Impact ⇒ S/C Structure
- Hostile Environment
  - Type, Level of Threat -> S/C Structure

10.1 Requirements, Constraints, and the Design Process

◆ Launch

- Launch Strategy
  - Single (Dual), Dedicated (Shared), Use of Upper Kick Stage (PKM, AKM) ⇒ S/C Orbit, Cost
- Boosted Weight
  - Launch Capabilities ⇒ S/C Weight ⇒ All Subsystem
- Envelope
  - Size, Shape ⇒ S/C Size, Shape
- Interfaces
  - Electrical, Mechanics ⇒ S/C Structure
- Launch Sites
  - Locations, Launch Azimuth ⇒ S/C Orbit

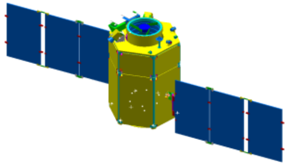
◆ Ground System Interface

- Degree of Autonomy
  - Reduce Ground Operation Cost ⇒ Required Autonomous Operations ⇒ S/C ADCS, C&DH
- Ground stations
  - Numbers, Locations, Performance ⇒ S/C Comm., C&DH, ADCS
- Space Links
  - Space-to-Space Link, Performance ⇒ S/C Comm., C&DH

10.1 Requirements, Constraints, and the Design Process

TABLE 10-4. Initial Spacecraft Design Decisions or Trade-offs. Further discussion of these trades is in Sec. 10.2.

Design Approach or Aspect	Where Discussed	Principal Options or Key Issues
Spacecraft Weight	Table 10-10	Must allow for spacecraft bus weight and payload weight.
Spacecraft Power	Tables 10-8, 10-9	Must meet power requirements of payload and bus.
Spacecraft Size	Sec. 10.1	Is there an item such as a payload antenna or optical system that dominates the spacecraft's physical size? Can the spacecraft be folded to fit within the booster diameter? Spacecraft size can be estimated from weight and power requirements.
Attitude Control Approach	Secs. 10.2, 10.4.2, 11.1	Options include no control, spin stabilization, or 3-axis control: selection of sensors and control torquers. Key issues are number of items to be controlled, accuracy, and amount of scanning or slewing required.
Solar Array Approach	Secs. 10.2, 10.4.6, 11.4	Options include planar, cylindrical, and omnidirectional arrays either body mounted or offset.
Kick Stage Use	Chaps. 17, 18	Use of a kick stage can raise injected weight. Options include solid and liquid stages.
Propulsion Approach	Secs. 10.2, 10.4.1, 17.2, 17.3	Is metered propulsion required? Options include no propulsion, compressed gas, liquid monopropellant or bipropellant.



10.1 Requirements, Constraints, and the Design Process

- **Spacecraft Weight**
  - S/C Bus Weight, Payload Weight ⇒ S/C Weight
- **Spacecraft Power**
  - S/C Bus Power, Payload Power ⇒ S/C Power
- **Spacecraft Size**
  - Payload Size, Booster Diameter, S/C Weight & Power ⇒ S/C Size
- **Attitude Control approach**
  - Items Controlled, Accuracy ⇒ Method of Power Supply ⇒ S/C Shape, Size, Weight
- **Solar Array Approach**
  - Power Required ⇒ Method of Power Supply ⇒ S/C Shape, Size, Weight
- **Kick stage use**
  - S/C Orbit ⇒ PKM, AKM ⇒ S/C Shape, Size, Weight
- **Propulsion Approach**
  - Orbit Control Required ⇒ Thruster (Propulsion) ⇒ S/C Shape, Size, Weight

10.2 Spacecraft Configuration

◆ Overview of S/C Design and Sizing (Table 10-1)

TABLE 10-1. Overview of Spacecraft Design and Sizing. The process is highly iterative, normally requiring several cycles through the table even for preliminary designs.

Step	References
1. Prepare list of design requirements and constraints	Sec. 10.1
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6. Iterate, negotiate, and update requirements, constraints, design budgets	Steps 1 to 5

10.2 Spacecraft Configuration

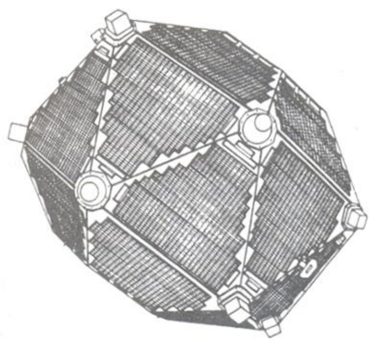
- **To estimate the size and structure of a S/C:**
  - Select design Approach
    - Top-Down Approach
      - Allocated design requirements are dictated by considering the overall S/C design.
    - Bottom-Up Approach
      - Allocated design requirements are dictated by gathering detailed design information.
  - Develop a S/C configuration (overall arrangement)
  - Make performance allocations to the S/C subsystem.

Different S/C Configuration Examples

- Equipment Compartment, Solar Array, Appendages, Attitude Control.

A. Spin-Stabilized Spacecraft

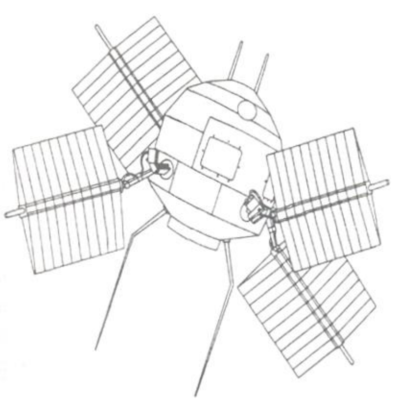
VELA



Mission: Nuclear Detection
Orbit: Super synchronous ~107,000 km
Payload: Radiation instruments body mounted
Configuration Features:
- *Equipment compartment:* 1.6 m diapolyhedron; internal solid AKM
- *Solar array:* Body mounted solar panels
- *Appendages:* None
- *Attitude control:* Spin-stabilized, inertially pointed

Weight: 221 kg *Power:* 90 W

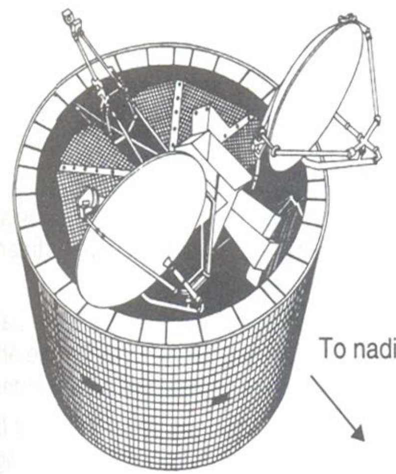
EXPLORER VI



Mission: Radiation Fields Measurement
Orbit: 245 × 42,400 km 47 deg inclination
Payload: Radiation Instruments body mounted
Configuration Features:
- *Equipment Compartment:* 0.6 m dia sphere
- *Solar Array:* Four deployed paddles 2.2 m span
- *Appendages:* Whip antennas
- *Attitude control:* Spin-stabilized, inertially pointed

Weight: 64.6 kg

DSCS II



Mission: Communications
Orbit: Geosynchronous
Payload: Communications transponder; Earth coverage horn antennas, and steerable pencil beam antennas
Configuration Features:
- *Equipment Compartment:* 3 m dia cylinder, 4.2 m long
- *Solar Array:* Body mounted on cylinder
- *Appendages:* Despun antenna platform with steerable parabolic antennas
- *Attitude control:* Spin-stabilized, spin axis normal to orbit plane

Weight: 523 kg
Power: 535 W BOL, 360 W EOL

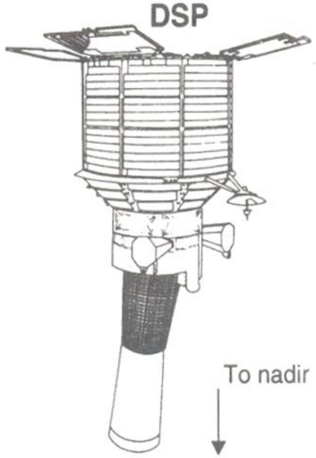
Fig. 10-1. **Typical Spacecraft Showing Different Configuration Options.** FOV = field-of-view; BOL = beginning-of-life; EOL = end-of-life.

Different S/C Configuration Examples

- Equipment Compartment, Solar Array, Appendages, Attitude Control.

B. 3-Axis-Stabilized Spacecraft

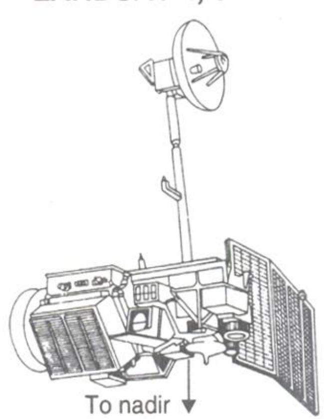
DSP



Mission: Earth Observation
Orbit: Geosynchronous
Payload: Body mounted telescope and attitude sensors
Configuration Features:
- *Equipment Compartment:* 4.5 m dia cylinder
- *Solar Array:* Body mounted on cylinder and on deployed panels
- *Appendages:* Solar panels and communications antennas
- *Attitude Control:* Cylinder axis pointed toward nadir; Body rotates slowly about cylinder axis to scan the payload FOV

Weight: 2,273 kg ***Power:*** 1,274 W

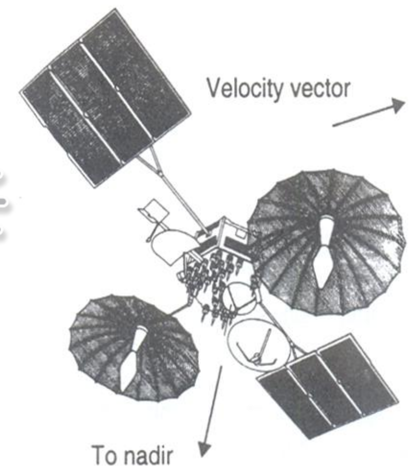
LANDSAT 4, 5



Mission: Earth Observation
Orbit: 700 km, Sun synchronous
Payload: Multispectral scanner fixed to spacecraft body with internal scanning mirror
Configuration Features:
- *Equipment Compartment:* Triangular cylinder 2 m dia, 4 m long
- *Solar Array:* Four panel deployed planar, single axis pointing control
- *Appendages:* Communication antenna
- *Attitude Control:* 3-axis control, 1 face toward nadir, 1 axis in direction of flight

Wt: 940 kg ***Power:*** 990 W BOL, 840 W EOL

TDRS



Mission: Communications
Orbit: Geosynchronous
Payload: S, K, and C band communication transponders with multiple antennas
Configuration Features:
- *Equipment Compartment:* 2.5 m hexagonal cylinder. Auxiliary compartments behind each of the large steerable antennas
- *Solar Array:* Deployed planar panels on both sides of equipment compartment. Single-axis articulation
- *Appendages:* Two 5 m steerable parabolic antennas, one 2 m steerable antenna, one 1.5 m fixed parabolic antenna
- *Attitude Control:* 3-axis control, one face toward nadir, 1-axis in direction of flight

Weight: 2,200 kg ***Power:*** 1,700 W

Fig. 10-1. Typical Spacecraft Showing Different Configuration Options. (Continued) FOV = field-of-view; BOL = beginning-of-life; EOL = end-of-life.

Different S/C Configuration Examples

- Equipment Compartment, Solar Array, Appendages, Attitude Control.

C. Spacecraft Configurations Featuring Long Booms

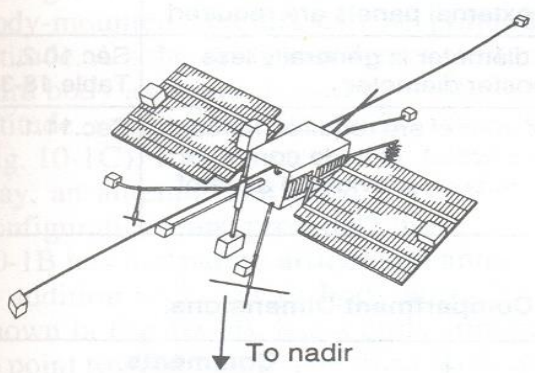
GEOSAT 	***Mission:*** Mapping the Earth's shape ***Payload:*** Radar altimeter ***Configuration Features:*** – *Solar Array:* Eight panels, each 3 m in length – *Appendages:* 7 m boom with 50 kg tip weight – *Attitude Control:* Gravity-gradient long axis vertical ***Weight:*** 636 kg
OGO 	***Mission:*** Observation of Geophysical Phenomena ***Payload:*** Multiple instruments for particle, fields and radiation measurements ***Configuration Features:*** – *Equipment Compartment:* Rectangular 1 m × 1 m × 2 m – *Solar Array:* Planar panels with single axis articulation – *Appendages:* Boom-mounted instruments, single-axis articulated instrument package – *Attitude control:* 3-axis Earth-Sun control using reaction wheels and cold gas thrusters ***Weight:*** 520 kg ***Power:*** 560 W

Fig. 10-1. Typical Spacecraft Showing Different Configuration Options. (Continued) FOV = field-of-view; BOL = beginning-of-life; EOL = end-of-life.

10.2 Spacecraft Configuration

TABLE 10-5. Spacecraft Configuration Drivers.

Configuration Driver	Effect	Rule of Thumb	Reference
Payload Weight	Spacecraft dry weight	Payload weight is between 17% and 50% of spacecraft dry weight. Average is 30%	Chap. 9
Payload Size and Shape	Spacecraft size	Spacecraft dimensions must accommodate payload dimensions	Chaps. 9, 10, 11, 13
Payload Power	Spacecraft power	Spacecraft power is equal to payload power plus an allowance for the spacecraft bus and battery recharging	Chap. 9, 10.2, 10.3
Spacecraft Weight	Spacecraft size	Spacecraft density will be between 20 kg/m^3 and 172 kg/m^3. Average is 79 kg/m^3	Secs. 10.2, 10.3
Spacecraft Power	Solar array area	The solar array will produce approximately 100 W/m^2 of projected area	Sec. 11.4
Solar Array Area	Solar array type	If required solar array area is larger than area available on equipment compartment, then external panels are required	Sec.11.4
Booster Diameter	Spacecraft diameter	Spacecraft diameter is generally less than the booster diameter	Sec.10.2, Table 18-3
Pointing Requirements	Spacecraft body orientation and number of articulated joints	Two axes of control are required for each article to be pointed. Attitude control of the spacecraft body provides 3 axes of control	Sec.11.1

7% Efficiency

Body-Mounted vs. Panel-Mounted

10.2 Spacecraft Configuration

- **Pointing Req's ⇒ S/C Body Orientation, Number of Articulated Joints**
  - S/C configuration must provide 2 axes of control for each item that is to be pointed
    - One body axis (i.e. the yaw axis) can be pointed toward nadir by control about the other 2 axes (roll & pitch)
      - Nadir Antenna: 2 axes (pitch, roll)
      - Solar array – 2 axes (yaw, a single axis)
  - Attitude Control Method ⇒ Solar Array Configuration
  - S/C configuration is driven to use long booms either to:
    - control S/C moment of inertia
    - separate delicate instruments from the S/C body electrical fields.

10.2 Spacecraft Configuration

TABLE 10-6. Estimating Spacecraft Equipment Compartment Dimensions.

Step	Procedure	Comments
1. Payload Weight	Starting point	Sec. 9.1
2. Estimate Spacecraft Dry Weight	Multiply payload weight by between 2 and 7	Average is 3.3
3. Estimate Spacecraft Propellant	Prepare a bottom-up propellant budget or arbitrarily select a weight	Normal range is 0% to 25% of spacecraft dry weight (Table 10-7)
4. Estimate Spacecraft Volume	Divide spacecraft loaded weight by estimated density	Range of density is 20–172 kg/m^3; Average is 79 kg/m^3
5. Select Equipment Compartment Shape and Dimensions	Shape and dimensions should match payload dimensions and fit within the booster diameter	In the folded configuration, spacecraft are cylindrically symmetric about the booster thrust axis. Cross-sectional shapes range from triangular to circular.

- Booster Diameter ⇒ S/C Diameter (+Known S/C Volume) ⇒ S/C Length